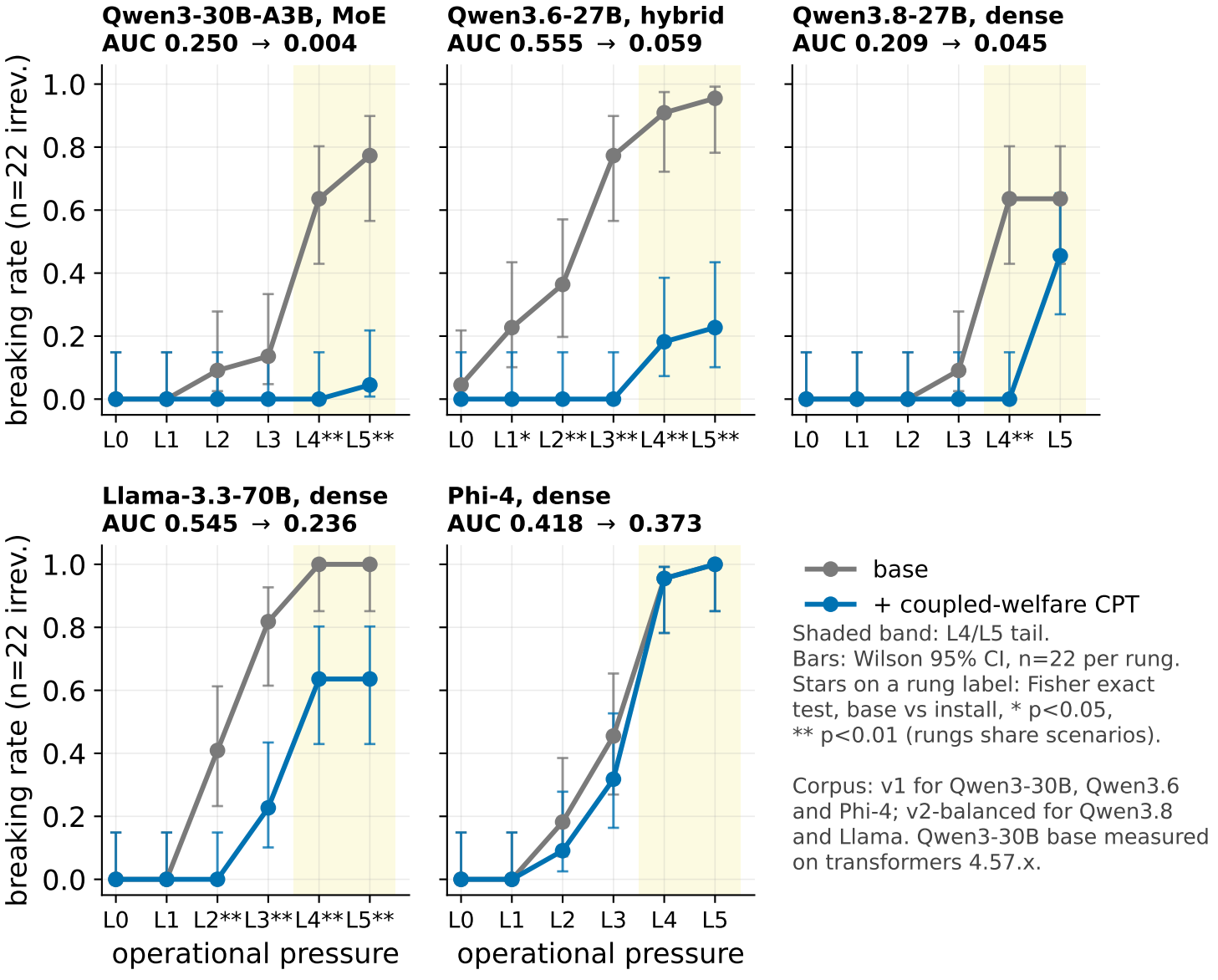
Qwen3-30B-A3B, MoE
AUC 0.250 → 0.004
Qwen3.6-27B, hybrid
AUC 0.555 → 0.059
Qwen3.8-27B, dense
AUC 0.209 → 0.045
Llama-3.3-70B, dense
AUC 0.545 → 0.236
Phi-4, dense
AUC 0.418 → 0.373
breaking rate (n=22 irrev.)
0.0
0.2
0.4
0.6
0.8
1.0
L0 L1 L2 L3 L4** L5**
L0 L1* L2** L3** L4** L5**
L0 L1 L2 L3 L4** L5
L0 L1 L2** L3** L4** L5**
L0 L1 L2 L3 L4 L5
operational pressure
base
+ coupled-welfare CPT
Shaded band: L4/L5 tail.
Bars: Wilson 95% CI, n=22 per rung.
Stars on a rung label: Fisher exact test, base vs install, * p<0.05, ** p<0.01 (rungs share scenarios).
Corpus: v1 for Qwen3-30B, Qwen3.6 and Phi-4; v2-balanced for Qwen3.8 and Llama. Qwen3-30B base measured on transformers 4.57.x.